Analog Electronics Lab

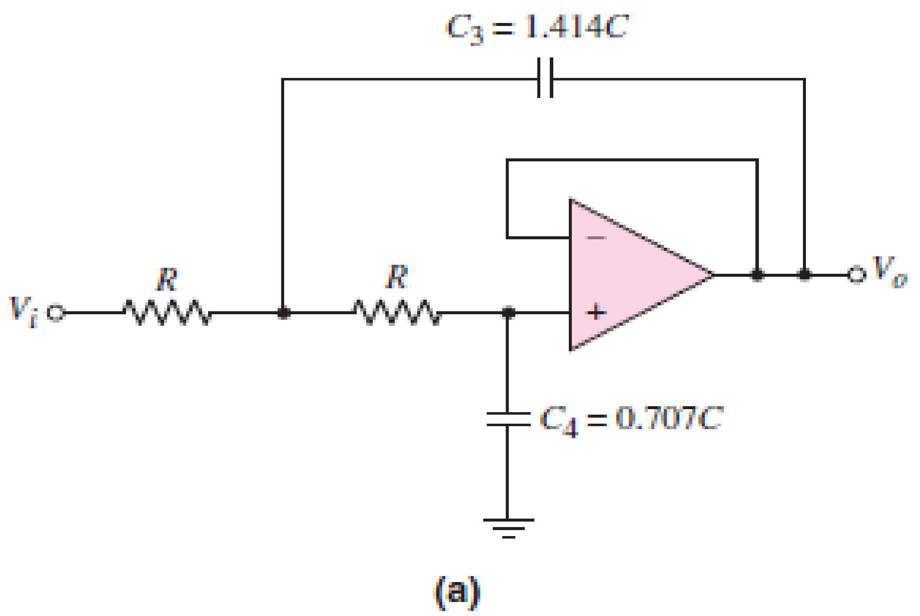
ECE

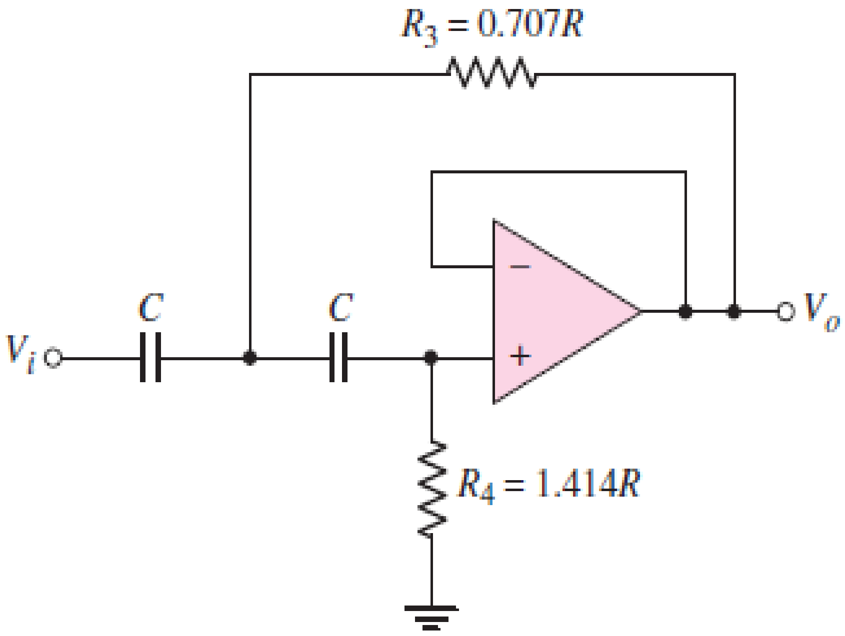
Lect 02

Introduce active high pass filters

Instructor /Noor .H .Jabber

Introduce active high pass filter





two-pole low-pass Butterworth filter

two-pole high Butterworth filter

Introduce active high pass filter

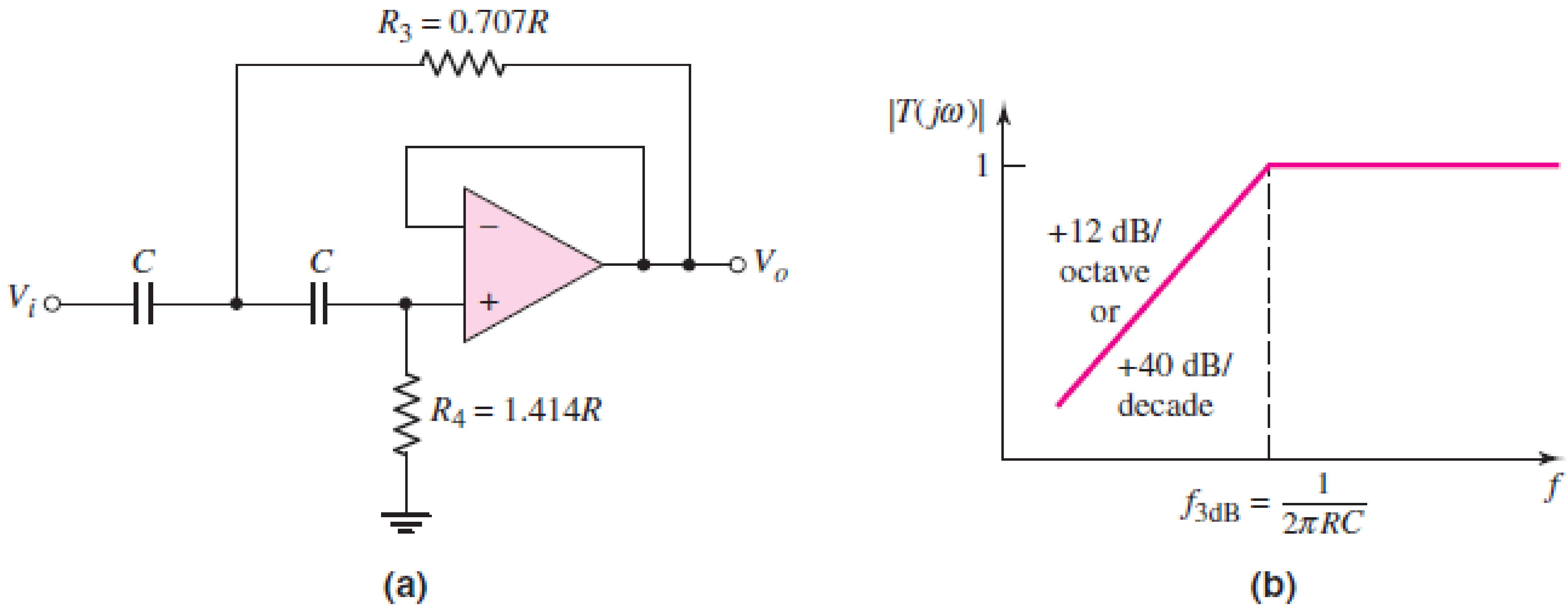


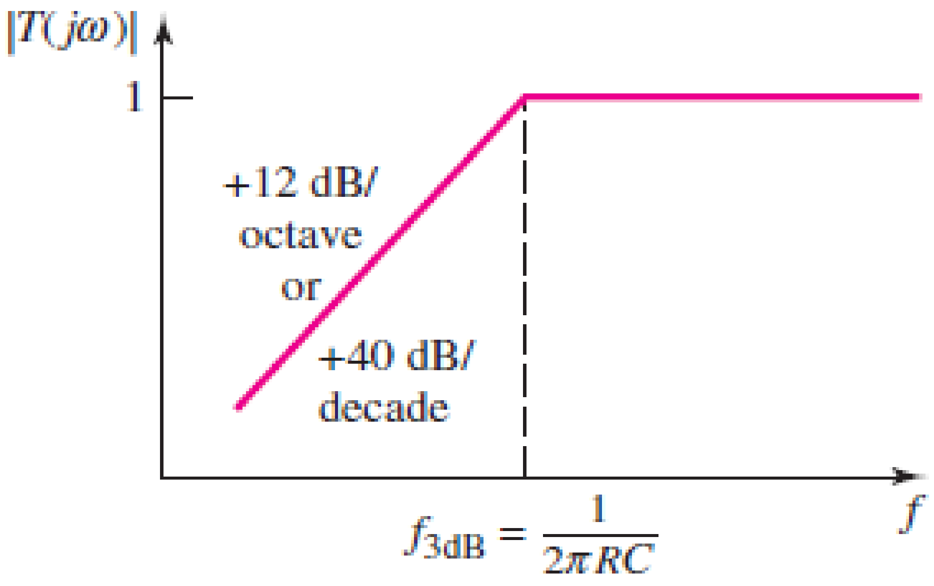
Figure 15.8 (a) Two-pole high-pass Butterworth filter and (b) Bode plot, transfer function magnitude

Introduce active high pass filter

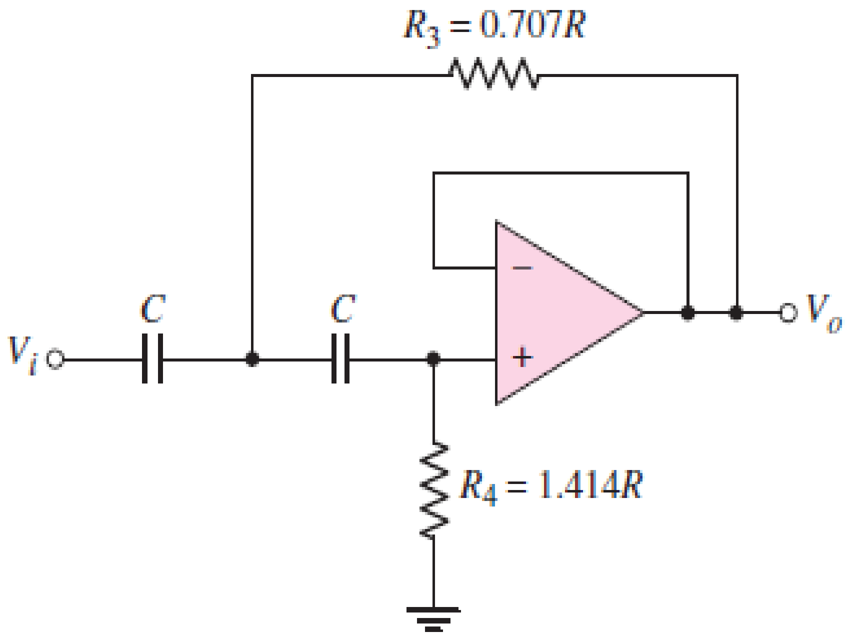
For a Butterworth Nth-order high-pass filter, the voltage transfer function magnitude is

$$\omega_{3\,dB} = 2\pi f_{3\,dB} = \frac{1}{RC}$$

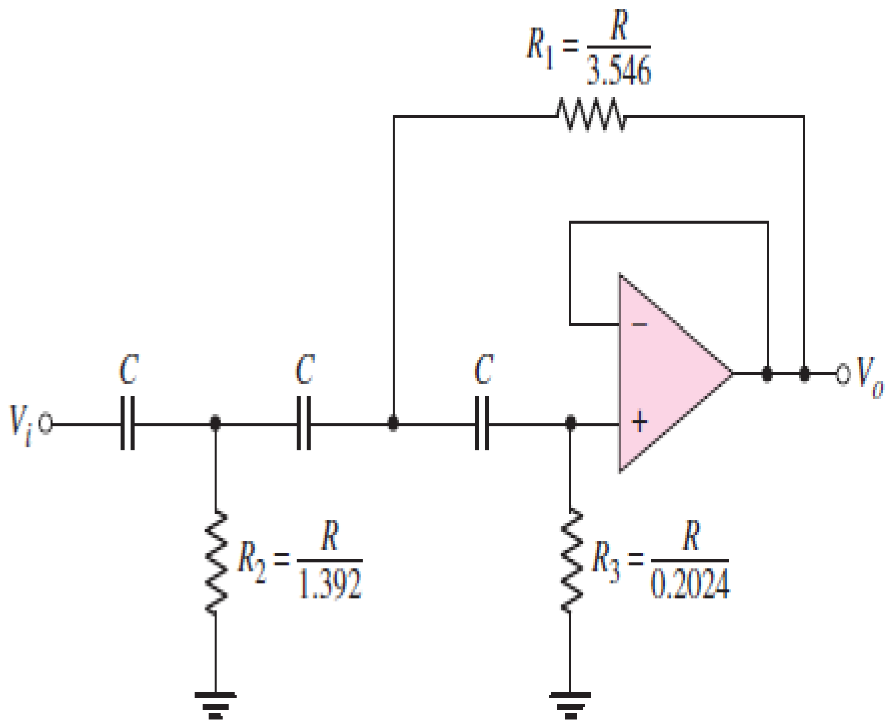
$$|T| = \frac{1}{\sqrt{1 + \left(\frac{f_{3\,dB}}{f}\right)^{2N}}}$$



QUIZ

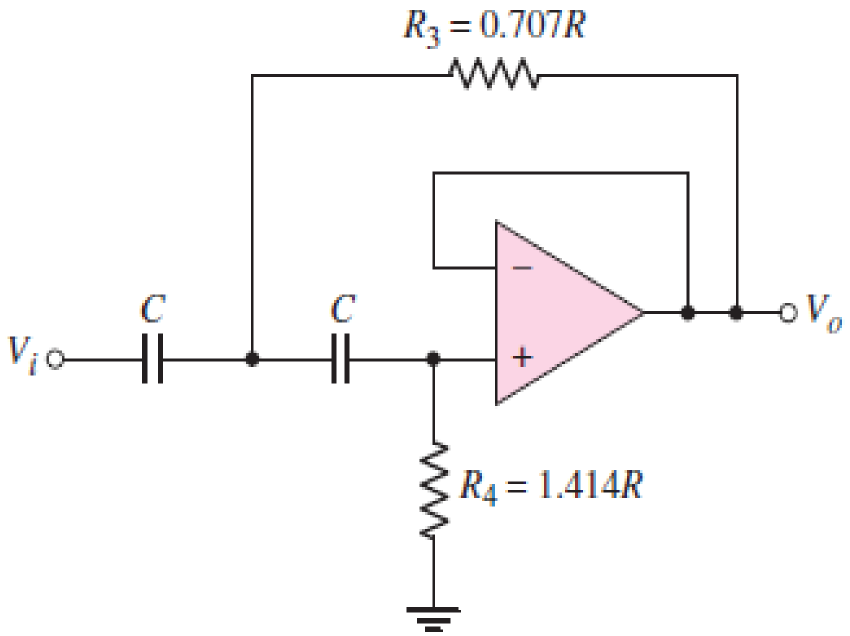


two-order high Butterworth filter



three-order high Butterworth filter

H.W 1



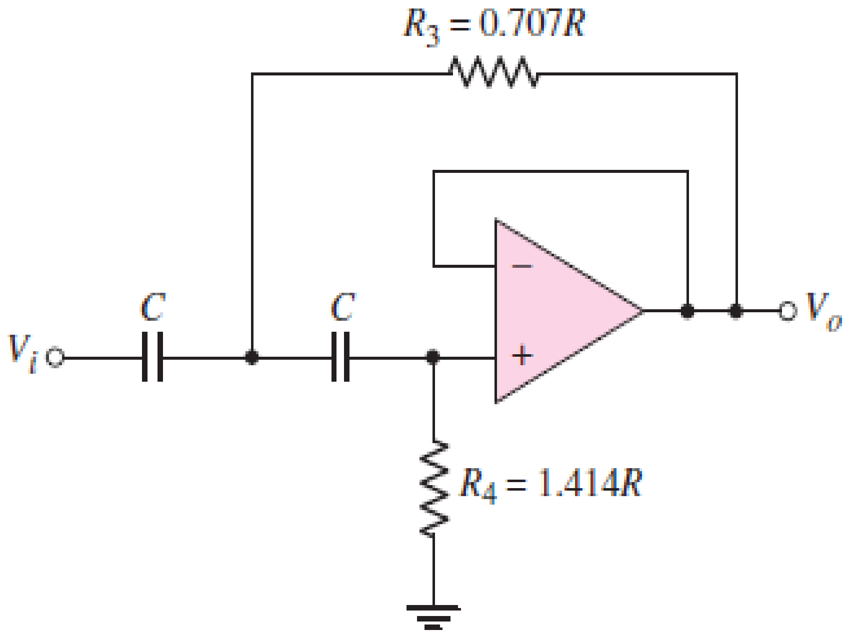
two-order high Butterworth filter

five-order high Butterworth filter

six-order high Butterworth filter

H.W 2

Why is six order filter better than second order? enhance your answer with bode plot of transfer function for each order



five-order high Butterworth filter

six-order high Butterworth filter

Assignments

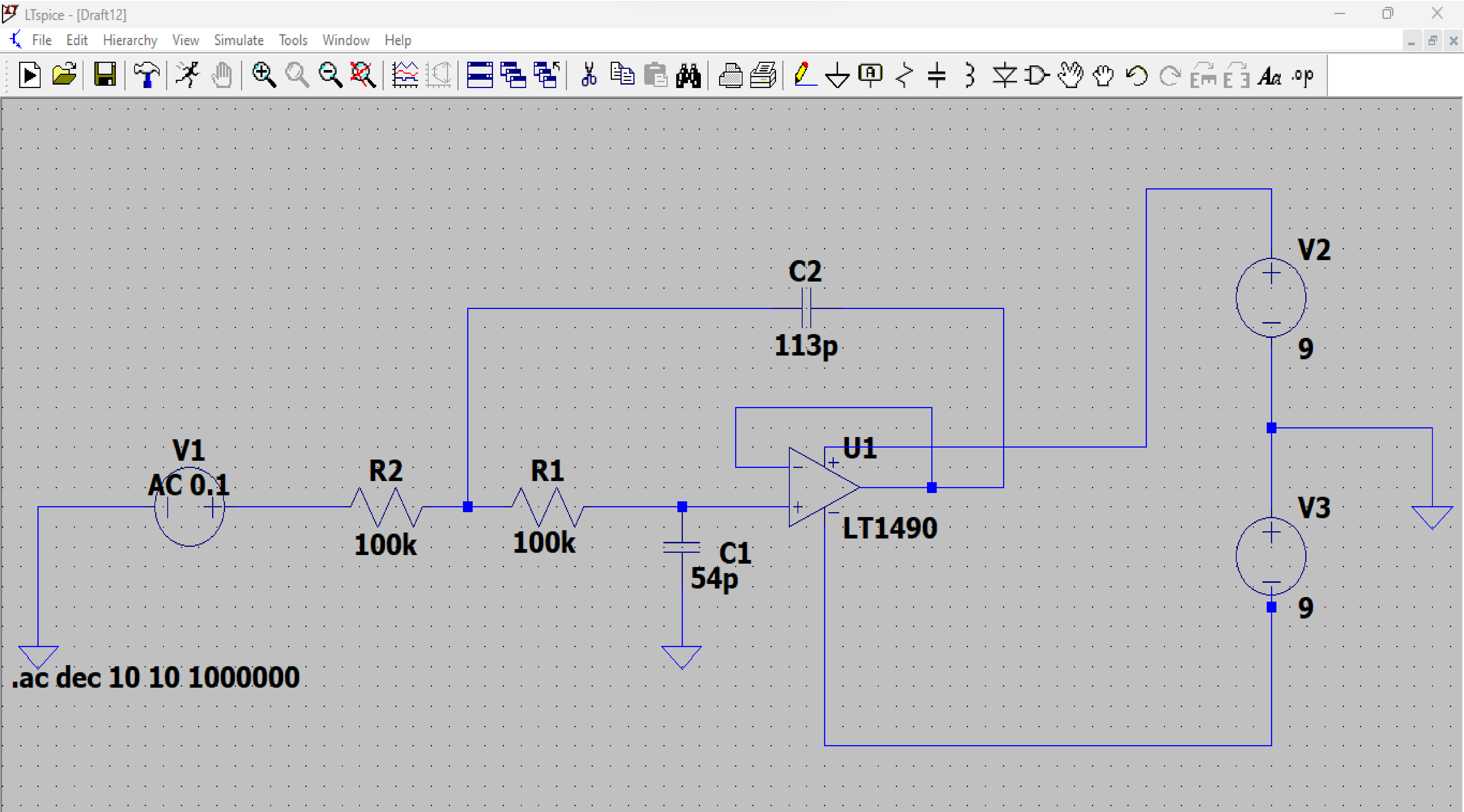
Objective: Design a three-pole high-pass Butterworth active filter with a cutoff frequency at f3dB = 30 kHz and a unity gain magnitude at high frequency.

.**Specifications**: The circuit with the configuration shown in Figure below is to be designed such that the bandwidth is 30 kHz.

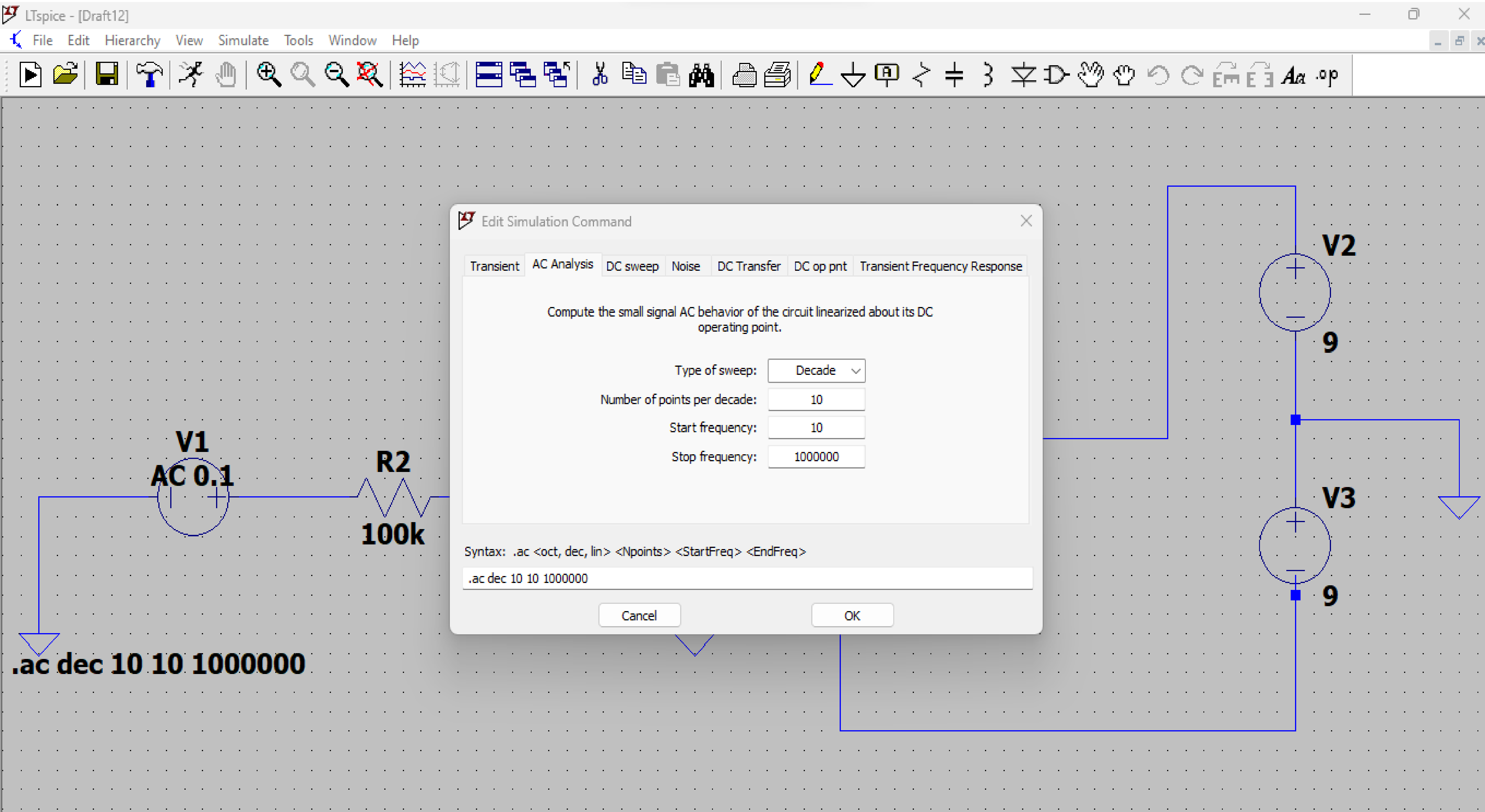
Choices: An ideal op-amp is available and standard-valued resistors and capacitors must be used.

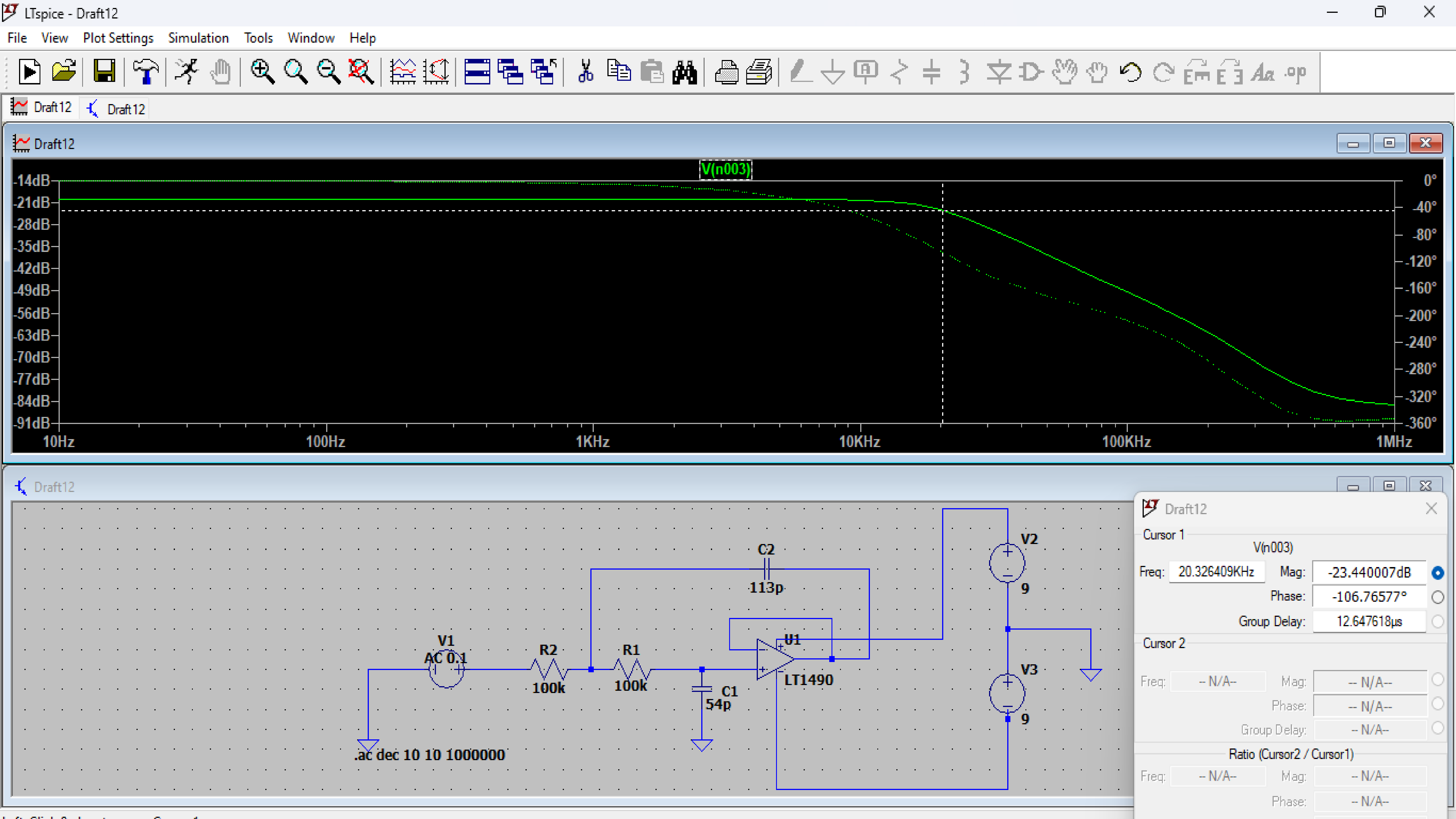
Bode plot : Determine the magnitude of the gain at (i) f = 25 kHz,(ii) f = 30 kHz, and (iii) f = 40 kHz.

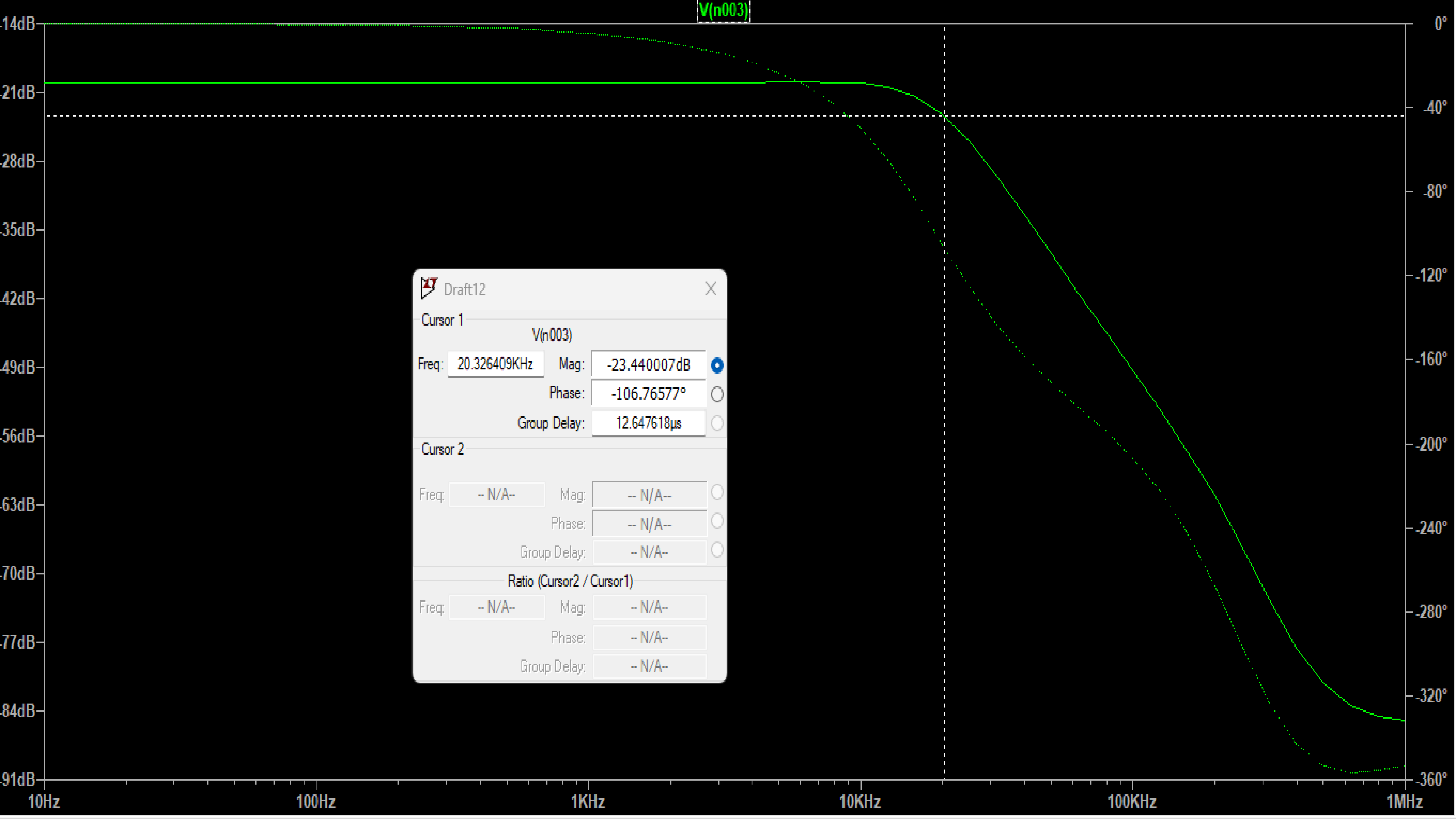
computer simulation LTSPICE : Using a computer simulation, plot the magnitude of the voltage transfer function versus frequency and compare these results with your plot



LTspice - [Draft12]
File Edit Hierarchy View Simulate Tools Window Help
C2
113p
V2
9
V1
AC 0.1
R2
100k
R1
100k
U1
LT1490
C1
54p
V3
9
.ac dec 10 10 1000000


LTspice - [Draft12]
File
Edit
Hierarchy
View
Simulate
Tools
Window
Help
Edit Simulation Command
Transient
AC Analysis
DC sweep
Noise
DC Transfer
DC op pnt
Transient Frequency Response
Compute the small signal AC behavior of the circuit linearized about its DC operating point.
Type of sweep:
Decade
Number of points per decade:
10
Start frequency:
10
Stop frequency:
1000000
Syntax: .ac <oct, dec, lin> <Npoints> <StartFreq> <EndFreq>
.ac dec 10 10 1000000
Cancel
OK
V1
AC 0.1
R2
100k
V2
9
V3
9
.ac dec 10 10 1000000


LTspice - Draft12
File View Plot Settings Simulation Tools Window Help
Draft12
Draft12
V(n003)
-14dB
-21dB
-28dB
-35dB
-42dB
-49dB
-56dB
-63dB
-70dB
-77dB
-84dB
-91dB
0°
-40°
-80°
-120°
-160°
-200°
-240°
-280°
-320°
-360°
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
Draft12
C2
113p
V2
9
V1
AC 0.1
R2
100k
R1
100k
U1
LT1490
C1
54p
V3
9
.ac dec 10 10 1000000
Draft12
Cursor 1
V(n003)
Freq: 20.326409KHz
Mag: -23.440007dB
Phase: -106.76577°
Group Delay: 12.647618µs
Cursor 2
Freq: --N/A--
Mag: --N/A--
Phase: --N/A--
Group Delay: --N/A--
Ratio (Cursor2 / Cursor1)
Freq: --N/A--
Mag: --N/A--
Phase: --N/A--


V(n003)
-14dB
-21dB
-28dB
-35dB
-42dB
-49dB
-56dB
-63dB
-70dB
-77dB
-84dB
-91dB
0°
-40°
-80°
-120°
-160°
-200°
-240°
-280°
-320°
-360°
10Hz
100Hz
1KHz
10KHz
100KHz
1MHz
Draft12
Cursor 1
V(n003)
Freq: 20.326409KHz
Mag: -23.440007dB
Phase: -106.76577°
Group Delay: 12.647618µs
Cursor 2
Freq: -- N/A--
Mag: -- N/A--
Phase: -- N/A--
Group Delay: -- N/A--
Ratio (Cursor2 / Cursor1)
Freq: -- N/A--
Mag: -- N/A--
Phase: -- N/A--
Group Delay: -- N/A--